

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Summative Assessment 2

Summative Assessment (SA-2)

Class: Class 10

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1-6: Networking, HTML/CSS, Python Programming, SQL Database
Management

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

a. What will be the output?

```
x = [1, 2, 3]
y = x
y.append(4)
print(x)
```

- a. [1, 2, 3]
- b. [1, 2, 3, 4]
- c. [4, 1, 2, 3]
- d. Error

b. Which SQL command is used to add a new column to an existing table?

- c. ADD COLUMN
- d. ALTER TABLE ... ADD
- e. INSERT COLUMN
- f. CREATE COLUMN

c. Which CSS property is used to make text bold?

- d. font-weight: bold
- e. text-style: bold
- f. font-style: bold
- g. text-weight: bold

d. What does try-except block do in Python?

- e. Ignores all errors
- f. Handles exceptions gracefully
- g. Terminates the program
- h. Creates error logs

e. Which HTML tag creates a hyperlink?

- f. <link>
- g. <a>
- h. <href>

i. <url>

f. Which SQL function returns the current date and time?

g. GETDATE()

h. NOW()

i. CURRENT_TIMESTAMP

j. Both B and C

g. What is the output?

```
s = "Hello World"
print(s.split()[1])
```

a. Hello

b. World

c. Hello World

d. Error

h. Which CSS display value makes an element invisible but still takes space?

i. display: none

j. visibility: hidden

k. opacity: 0

l. display: invisible

i. In Python, which keyword is used for exception re-raising?

j. raise

k. throw

l. except

m. catch

j. Which SQL query displays the structure of a table?

k. SHOW TABLE

l. DESCRIBE table_name

m. STRUCTURE table_name

n. INFO table_name

Part B: Short Answer (8 x 2 = 16 marks)

a. Write Python code to read a CSV file containing student records (Name, Roll, Marks) and display the top 3 scorers. Use the csv module.

b. Write the output of the following and explain each line:

```
import json
data = {"students": [{"name": "A", "marks": 85}, {"name": "B", "marks": 92}]}
json_str = json.dumps(data, indent=2)
print(json_str)
```

```
parsed = json.loads(json_str)
print(parsed["students"][1]["name"])
```

- c. Write SQL queries for an **Online Shopping** database:
- Customers(cid, name, email, city)
 - Products(pid, name, price, stock)
 - Orders(oid, cid, pid, quantity, order_date)
- (a) Find customers from 'Bangalore' (b) List products with stock less than 10 (c) Find total orders placed by each customer (d) Find the most ordered product
- d. Differentiate between list and tuple in Python with 5 points. When would you use each?
- e. Write CSS code to create a **pricing card** component with:
- Card with shadow and rounded corners
 - Header with gradient background
 - Price display in large font
 - Feature list with checkmark icons
 - Call-to-action button with hover effect
- f. Explain **normalization** in databases. What is the difference between 1NF, 2NF, and 3NF? Give an example table that violates each.
- g. Write a Python function `merge_sort(arr)` that implements the merge sort algorithm. Include proper comments and handle edge cases.
- h. What is the purpose of JOIN in SQL? Explain the difference between INNER JOIN, LEFT JOIN, and RIGHT JOIN with diagrams and examples.

Part C: Long Answer (4 x 3 = 12 marks)

- a. Write a complete Python program for a **Library Management System** that:
- Uses file handling to store book records (title, author, ISBN, available_copies)
 - Implements functions for: add_book, search_book, issue_book, return_book, display_all
 - Handles exceptions for file operations and invalid inputs
 - Generates a daily report of issued and returned books

Board Exam Pattern - Karnataka 2024

- b. Create a complete HTML/CSS webpage for a **Healthcare Appointment System** that includes:
- Header with hospital name and navigation
 - Appointment booking form (doctor selection, date, time, patient details)
 - Styled table showing available doctors with specialisation and timings
 - Responsive design with media queries
 - Professional medical-themed colour scheme

Previous Year Pattern - Karnataka Board 2023

- c. Write SQL queries for a **School Examination System** with tables:
- Students(roll, name, class, section)
 - Subjects(sub_id, sub_name, max_marks)
 - Marks(roll, sub_id, marks_obtained)
- (a) Create a **result card** showing each student's marks, total, and percentage (b) Find the class topper (c) Find subject-wise toppers (d) List students who failed in any subject (marks < 35% of max_marks) (e) Find the overall pass percentage (f) Display the rank list with rank, name, total, and percentage

- d. Explain the complete process of **data communication** from one computer to another. Include:
- Components of data communication (sender, receiver, medium, protocol)
 - Types of transmission (simplex, duplex, multiplex)
 - Errors in data transmission and how they are detected (parity check, CRC)
 - Role of TCP in reliable data transfer with a diagram

Part D: Application Based (2 x 1 = 2 marks)

- a. A student writes the following Python code to read a file:

```
f = open("data.txt")
content = f.read()
print(content)
f.close()
```

The program crashes if the file doesn't exist. Identify the issue and rewrite the code using try-except and with statement. Also explain why with is preferred.

- b. A database has the following Students table:

roll	name	class	section	marks
1	Rahul	10	A	85
2	Priya	10	A	92
3	Amit	10	B	78

Write the SQL query that produces this output:

section	avg_marks	topper
A	88.5	Priya
B	78.0	Amit

Explain each clause of your query.